

Some important sample papers of iT project management

Total Marks: 60

Time Allowed: 2 Hours

Instructions:

- Attempt all questions.
 - Be precise and to the point in your answers.
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Section A: Short Answer Questions

(Attempt all questions, each carries 5 marks.)

1. Define **Project Management** and explain its importance in **Information Technology**.
 2. What is the **Scope Triangle** in project management? Explain briefly.
 3. Explain the concept of **Work Breakdown Structure (WBS)** with an example.
 4. Describe the **five phases of the Project Management Life Cycle**.
 5. What is **Critical Path Method (CPM)**? How is it useful in project planning?
 6. Define **Earned Value Analysis (EVA)** and mention its key components.
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Section B: Long Answer Questions

(Attempt any three questions, each carries 10 marks.)

7. Explain the **Project Management Process Groups** and their significance in managing IT projects.
8. Discuss the **five methods of Estimating Activity Duration** with real-life examples.
9. Describe **PERT (Project Evaluation and Review Technique)**. How does it differ from CPM?
10. What are the different **team structures** used in software project management? Compare **Democratic Decentralized, Controlled Decentralized, and Controlled Centralized** structures.
11. What is **Risk Identification and Risk Management** in project management? Explain with examples.

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Instructions:

- Attempt all questions.
 - Be precise and to the point in your answers.
 - Show all necessary calculations where required.
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Section A: Short Answer Questions

(Attempt all questions, each carries 5 marks.)

1. Define **Project Management** and explain its importance in **Information Technology**.
 2. What is the **Scope Triangle** in project management? Explain briefly.
 3. Explain the concept of **Work Breakdown Structure (WBS)** with an example.
 4. Describe the **five phases of the Project Management Life Cycle**.
 5. What is **Critical Path Method (CPM)**? How is it useful in project planning?
 6. Define **Earned Value Analysis (EVA)** and calculate **Cost Performance Index (CPI)** given:
 - **Planned Value (PV) = \$50,000**
 - **Earned Value (EV) = \$40,000**
 - **Actual Cost (AC) = \$45,000**
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Section B: Long Answer Questions

(Attempt any three questions, each carries 10 marks.)

7. Explain the **Project Management Process Groups** and their significance in managing IT projects.
8. Discuss the **five methods of Estimating Activity Duration** with real-life examples. Also, calculate the **expected time (TE)** using **PERT formula** where:
 - **Optimistic Time (O) = 10 days**
 - **Most Likely Time (M) = 15 days**
 - **Pessimistic Time (P) = 20 days**
9. Describe **PERT (Project Evaluation and Review Technique)**. Calculate the **variance and standard deviation** for an activity with:
 - **O = 8 days, M = 12 days, P = 16 days.**
10. What are the different **team structures** used in software project management? Compare **Democratic Decentralized, Controlled Decentralized, and Controlled Centralized** structures.

11. What is **Risk Identification and Risk Management** in project management? Explain with examples. Also, determine the **risk exposure (RE)** given:

- **Probability of risk occurrence = 0.3**
- **Impact cost = \$20,000**

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Q.1 Short Questions (5 × 6 = 30 Marks)

Answer the following short questions.

1. Define Project Management and explain its importance in the IT industry.
2. What is the Scope Triangle in project management? Briefly explain its components.
3. Explain the concept of Earned Value Analysis (EVA) in project monitoring.
4. Differentiate between CPM (Critical Path Method) and PERT (Project Evaluation Review Technique).
5. What are the five phases of the Project Management Life Cycle?
6. How does Maslow's Need Theory apply to project team motivation?

Q.2 Long Questions (3 × 10 = 30 Marks)

1. **(Numerical Question)** A project has the following activities with estimated durations:

Activity	Predecessor	Duration (Days)
A	None	5
B	A	3
C	A	4
D	B	6
E	C	2
F	D, E	5

2. (a) Draw the project network diagram.
(b) Identify the Critical Path and calculate the project duration.
3. **(Numerical Question)** A project has a budget of \$50,000 and is expected to be completed in 5 months. After 3 months:
 - Planned Value (PV) = \$30,000
 - Earned Value (EV) = \$25,000

- Actual Cost (AC) = \$28,000
 - (a) Calculate Schedule Variance (SV) and Cost Variance (CV).
 - (b) Calculate Cost Performance Index (CPI) and Schedule Performance Index (SPI).
 - (c) Interpret the results and explain the project's current status.
4. Explain the concept of Change Management and its importance in project success. Discuss various baselines and how version control helps in project management.

